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Potential Cost Savings in the Reversal of DOAC Associated Bleeding



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Background

- Direct oral anticoagulants (DOACs) are a cornerstone of therapy in treatment and prevention of deep vein thrombosis, pulmonary embolism and other thromboembolic disorders
- Despite their favorable safety profile, patients are still at an increased risk of intracranial hemorrhages (ICH), gastrointestinal (GI) and/or fatal bleeding
- Two agents commonly used in DOAC associated bleeding include andexanet-alfa and four factor prothrombin concentrate complex (4F-PCC)
- Reversal agent selection is determined by clinical presentation, cost, and institutional preference
- 4F-PCC costs \$9,099 per patient compared to \$12,310 and \$22,158 for low dose and high dose of andexanet-alfa respectively

Objectives

- To evaluate adherence to Henry Ford Health System (HFHS) 2023 “Tier 1: Anticoagulation Reversal Guidelines” and determine cost efficiency in the management of DOAC-associated bleeding with andexanet-alfa

Methods

- Single-center, retrospective, cohort study that included patients admitted to a HFHS hospital and received andexanet-alfa between June 2022 and June 2023
- Inclusion Criteria: (1) 18 years or older, (2) on apixaban or rivaroxaban at home and (3) had a bleeding event requiring reversal of DOAC with andexanet-alfa
- Patients were grouped based on whether they appropriately or inappropriately received andexanet alfa according to HFHS anticoagulant reversal guideline(s)

Primary Endpoint

- Analyze the percentage of patients who qualified for prothrombin complex concentrate

Secondary Endpoint

- Determine the total cost savings if the 2023 guidelines had been followed

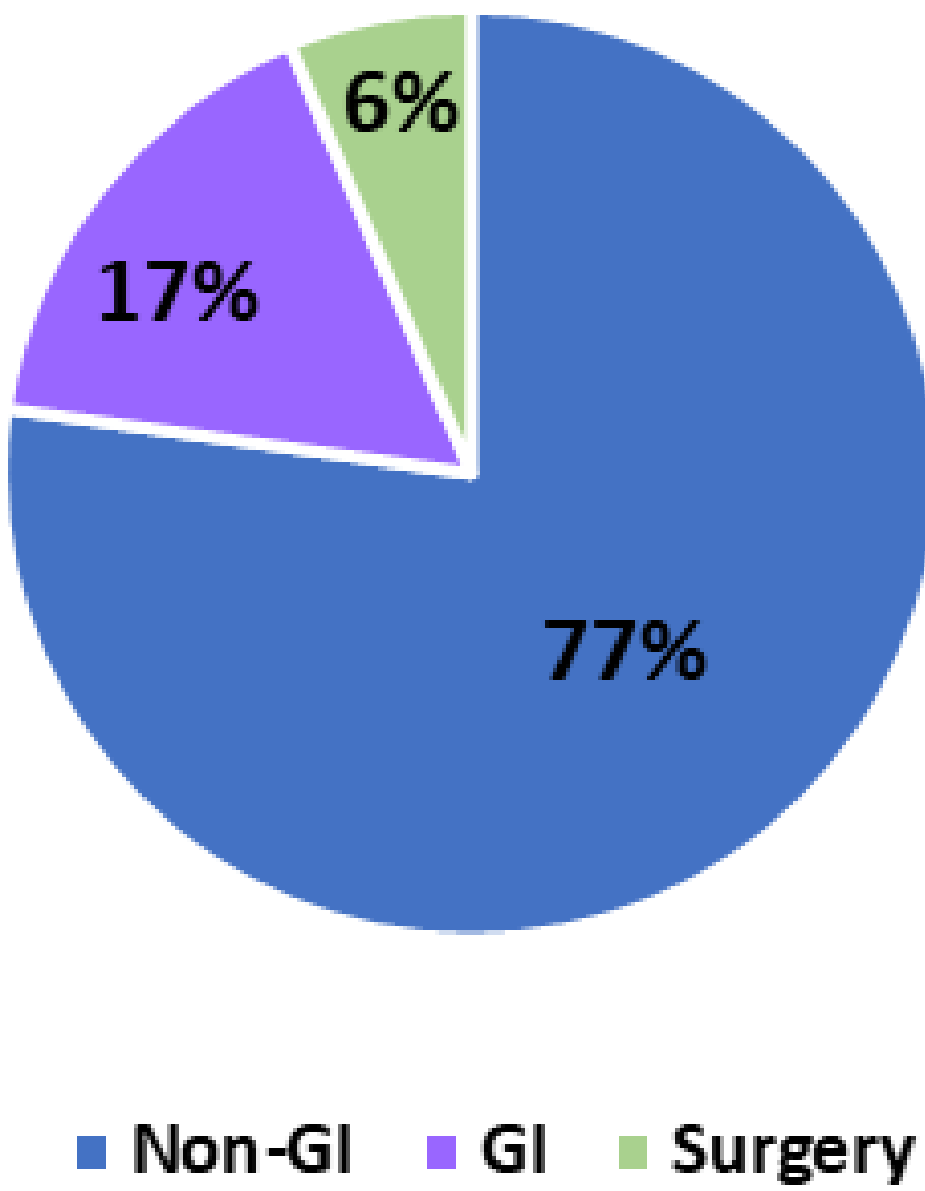
Table 1. Case Characteristics

Characteristics ^a	Total (n = 62)	Qualified for PCC (n = 13)	Did Not Qualify for PCC (n = 49)
Age [years], mean (SD)	74.5	73 (11.8)	76 (13.1)
Gender			
Male	38	8 (61.5%)	30 (61.2%)
Female	24	5 (38.5%)	19 (38.8%)
Race			
African American	4	3 (23.1%)	1 (2%)
Asian	0	0 (0%)	0 (0%)
Caucasian	58	10 (76.9%)	48 (98%)
Hispanic	0	0 (0%)	0 (0%)
Other	0	0 (0%)	0 (0%)
BMI (kg/m ²)	30.5	33.2 (9.2)	27.9 (7.1)
Baseline HgB (g/dL)	10.9	10.1 (3.5)	11.7 (2.5)
Baseline Platelets (plts/mcL)	246.5	272 (130)	221(99)
Length of Stay (days)	6.3	5.7 (4.2)	6.9 (5.8)
Baseline Blood Pressure	122/68	112/63 (31/19)	133/74 (31/16)
Dose of Andexanet-Alfa			
Low-Dose Andexanet-Alfa	59	13 (100%)	46 (93.9%)
High-Dose Andexanet-Alfa	3	0 (0%)	3 (100%)
Prior to Admission DOAC			
Apixaban	46	8 (61.5%)	37 (75.5)
Rivaroxaban	16	5 (38.5%)	11 (22.4%)

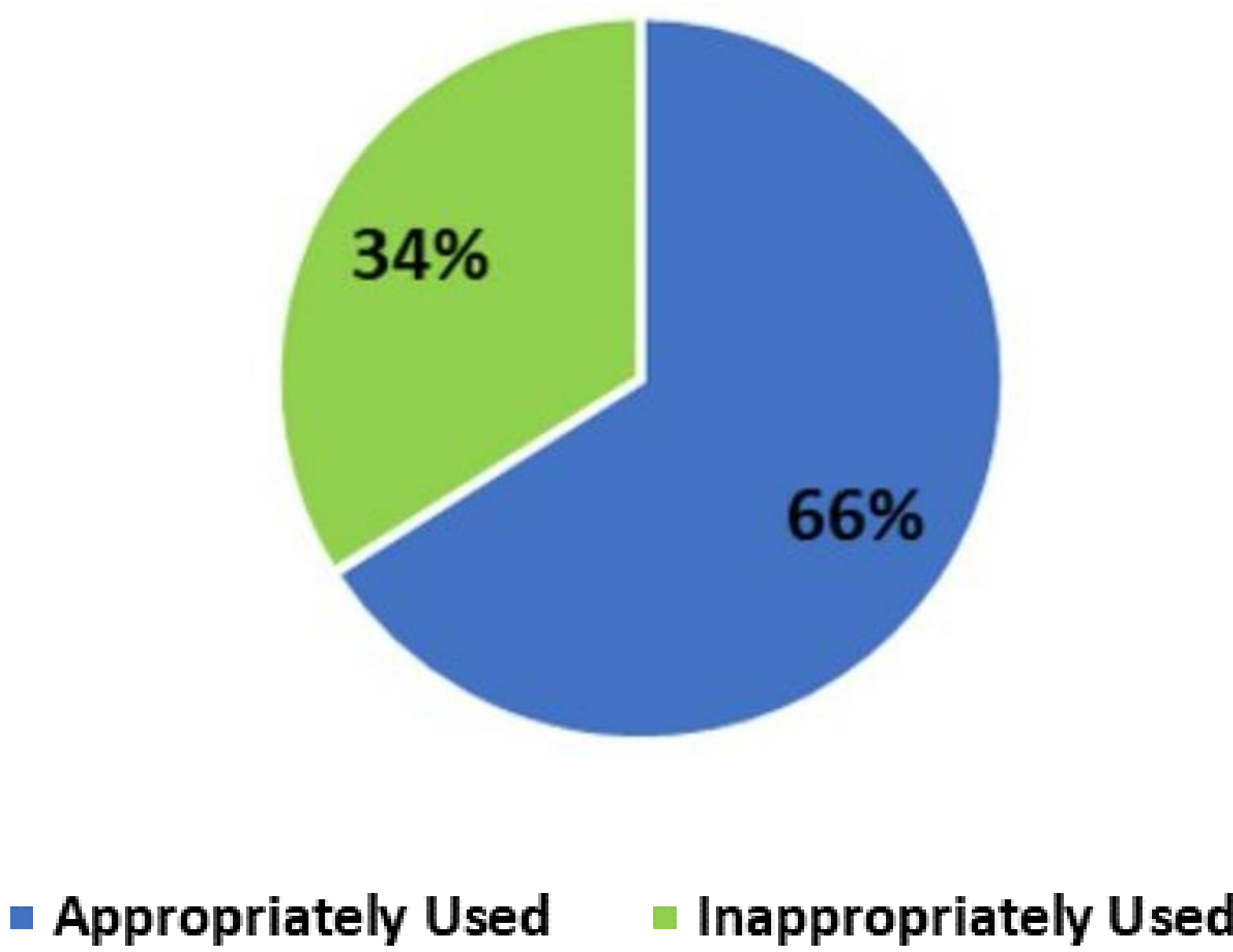
^aData are presented as “number (%)” or “mean (standard deviation)”, as appropriate
Abbreviations: BMI, body mass index; DOAC, direct oral anticoagulants

Results

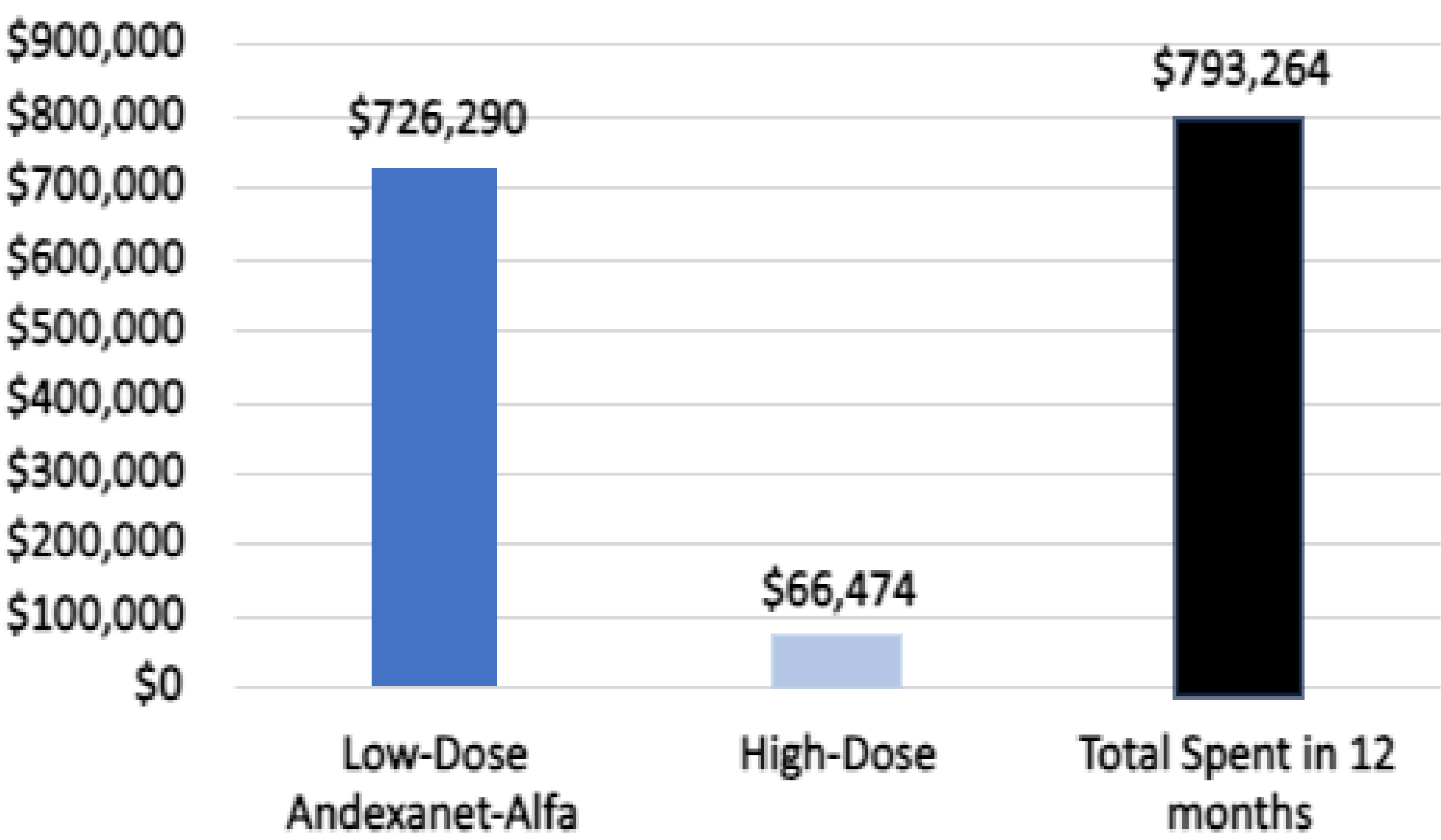
Andexanet-Alpha Indication



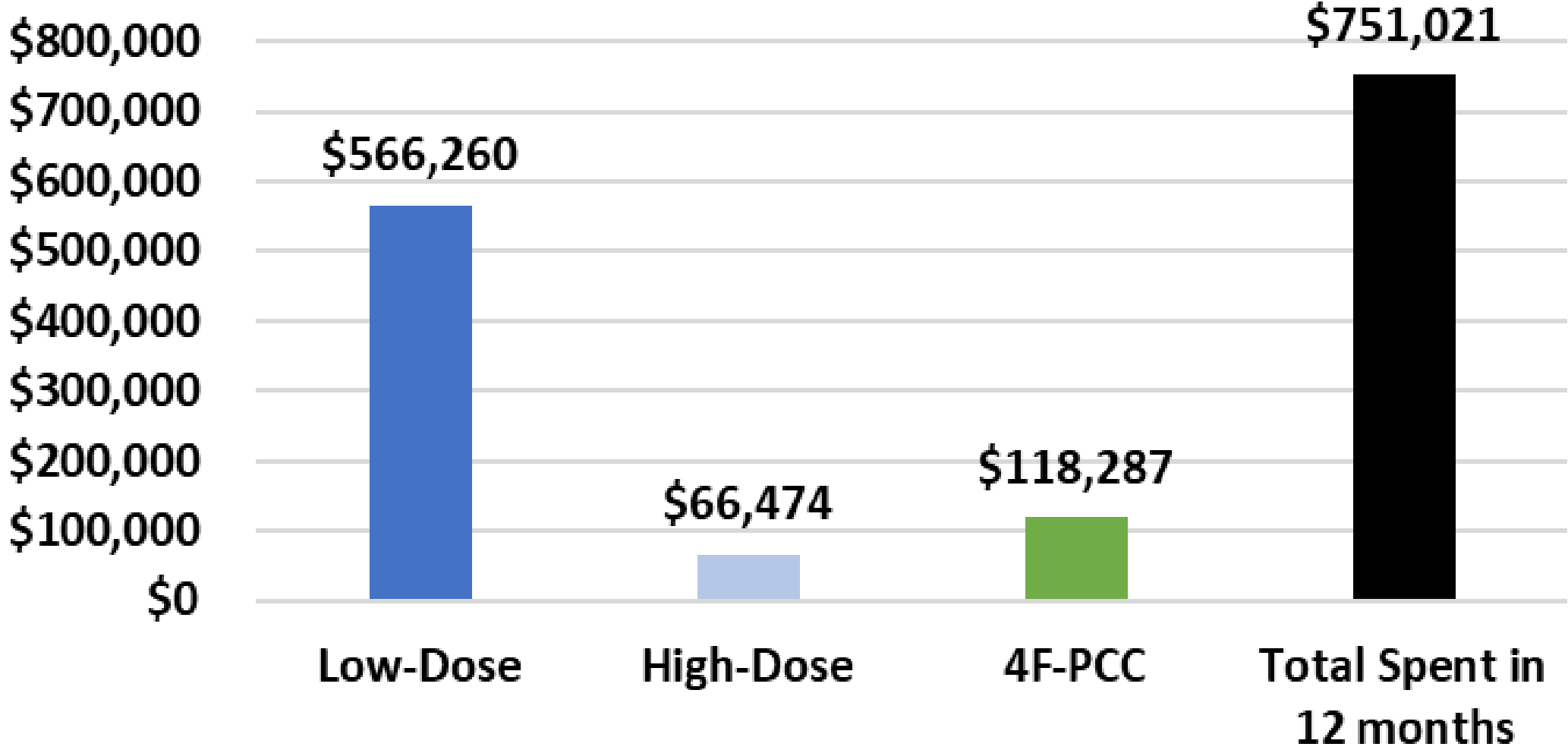
Andexanet-Alfa Use



Amount Spent on Andexanet-Alfa in 12 Months



Amount Spent if 2023 Guidelines Implemented in 12 Months



Conclusion

The Tier 1 Anticoagulation Reversal guideline was appropriately followed for 66% of the sampled assessed when using andexanet-alfa.

In patients with DOAC-associated GI bleeds, 4F-PCC is an effective and economical reversal agent with significant cost-savings opportunities by following the 2023 anticoagulation reversal guidelines at a HFHS.

The estimated total net savings from using 4F-PCC as an alternative to andexanet-alfa based upon the 2023 HFHS guidelines is \$42,243.

Further studies could be warranted to assess andexanet-alfa use in non-GI-associated bleeds to potentiate the cost-saving opportunities.

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4. Stevens VM, Trujillo TC, et al. *Clin Appl Thromb Hemost.* 2021; 27:10.1177.